



Module Title

Cybersecurity and Networking

Internship Assignment Submission

**Day 08 Challenge Submission — Tasks 1, 4 & 5 (Investigation
Evidence)**

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Role: Cybersecurity Intern

Duration: 3 months

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Day:7 (Seven)

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1. Practical Tasks

1.1 TASK 1 — NETWORK IDENTITY From ipconfig /all on YOUR computer record: IPv4 Address · Subnet Mask · Default Gateway · DNS Server · MAC/Physical Address · DHCP Enabled (Yes/No).

IPv4 Address. : 192.168.101.9(Preferred)

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.101.1

Physical Address. : 1C-CE-51-F4-2D-52

DNS Servers : 110.44.112.245

110.44.113.245

DHCP Enabled. : Yes

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . : 
Description . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
Physical Address. . . . . : 1C-CE-51-F4-2D-52
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2404:7c00:48:1e04:1e12:e394:1ac:9ee4(Preferred)
Temporary IPv6 Address. . . . . : 2404:7c00:48:1e04:f980:8667:d4fb:dd86(Preferred)
Link-local IPv6 Address . . . . . : fe80::5305:27d1:a329:219d%5(Preferred)
IPv4 Address. . . . . : 192.168.101.9(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Sunday, September 13, 2026 9:17:20 PM
Lease Expires . . . . . : Wednesday, September 16, 2026 2:28:55 PM
Default Gateway . . . . . : fe80::72a5:6aff:feb3:259a%5
                             192.168.101.1
DHCP Server . . . . . : 192.168.101.1
DHCPv6 IAID . . . . . : 85773905
DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-C2-C4-6A-30-13-8B-85-80-C6
DNS Servers . . . . . : fe80::1%5
                             110.44.112.245
                             110.44.113.245
                             fe80::1%5
NetBIOS over Tcpip. . . . . : Enabled
```

1.2 TASK 2 — NETWORK LAYER MATCHING (15 POINTS)

Quiz History

Day 08 Challenge — Tasks 2 & 3 (OSI Matchin...

9/15/2026, 2:51:55 PM

30/30

Passed

1.3 TASK 3 — IP ADDRESS CLASSIFICATION (15 POINTS)

Quiz History

Day 08 Challenge — Tasks 2 & 3 (OSI Matchin...

9/15/2026, 2:51:55 PM

30/30

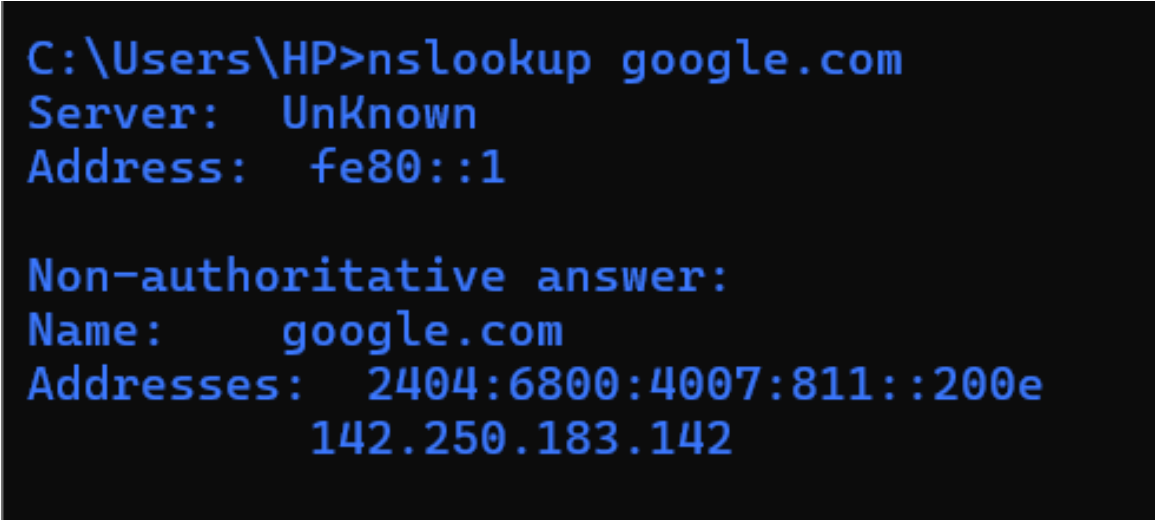
Passed

1.4 TASK 4 — DNS INVESTIGATION (6 marks) Run nslookup google.com. State: which DNS server responded, which IP address was returned, and what the purpose of DNS is. Attach a screenshot.

DNS Server: Unknown (fe80::1)

IP Address returned: 142.250.183.142

Purpose of DNS: DNS translates domain names such as google.com into IP addresses so computers can locate and connect to websites.



```
C:\Users\HP>nslookup google.com
Server:      UnKnown
Address:     fe80::1

Non-authoritative answer:
Name:        google.com
Addresses:   2404:6800:4007:811::200e
             142.250.183.142
```

1.5 TASK 5 — NETWORK PATH INVESTIGATION (6 marks) Run `tracert google.com`. State: how many visible hops appeared, what the first hop was, what device commonly represents the first hop, and what `tracert` is used for. Attach a screenshot.

First hop: 2404:7c00:40:5: e3:13e4

First-hop device: Usually the local router/ default gateway

Visible hops appeared: 13

Traceroute purpose: Traceroute shows the path packets take from your computer to a destination and helps identify network delays or connectivity problems.

```
C:\Users\HP>tracert google.com

Tracing route to google.com [2404:6800:4007:839::200e]
over a maximum of 30 hops:

  1    11 ms     6 ms     7 ms  2404:7c00:40:5::e3:13e4
  2    16 ms     2 ms     2 ms  2404:7c00::12
  3    19 ms     3 ms     6 ms  2404:7c00:0:d::1
  4    10 ms     3 ms     3 ms  2404:7c00:0:4007::1
  5    13 ms     3 ms     3 ms  2404:7c00:0:4002::1
  6    26 ms    66 ms     6 ms  2404:7c00:0:3::b
  7     7 ms     7 ms     7 ms  2404:a800:1a00:807::89
  8    65 ms    100 ms    102 ms 2404:a800::92
  9   109 ms     84 ms    101 ms 2404:6800:8201::1
 10    67 ms    101 ms    101 ms 2404:6800:8201::1
 11   104 ms    105 ms    101 ms 2001:4860:0:1::55ca
 12   114 ms    101 ms    101 ms 2001:4860:0:1::f3f
 13    57 ms     57 ms    100 ms lcmaaa-am-in-x0e.1e100.net [2404:6800:4007:8
39::200e]

Trace complete.
```

1.6 TASK 6 — FINAL FLAG CHALLENGE (20 POINTS)

NETWORK SECURITY

✓ Task 6 — Final Flag: Network Address Calculation

20 pts

Network Security

beginner

Scenario: a system has IPv4 192.168.50.25, subnet mask 255.255.255.0, default gateway 192.168.50.1, MAC AA-BB-CC-DD-EE-FF. Work out the network address, broadcast address, first usable host and last usable host. The flag is built from the NETWORK address: wrap it as JVL{...} replacing every dot with an underscore. Correct submission = FLAG CAPTURED (20 points). Type the flag EXACTLY: capital JVL, underscores between the numbers, no spaces.

💡 Show hint

🏆 Solved! Flag captured.

Cyber Security

Given

IP address: 192.168.50.25

Subnet mask: 255.255.255.0

Default Gateway: 192.168.50.1

So,

In a 124, the first 3 octets are network portion and last octet is host portion.

192.168.50.25 → Network Address 192.168.50.0

Now, for 124

Broadcast Address is .255

192.168.50.255

First Usable Host = +1

= 192.168.50.1

Last Usable Host = -1

= 192.168.50.254

Network Address = JVL 192-168-50-02